Initial Project Planning

UniClass AI: A University Learning Management System with an Explainable AI Student-Risk Advisor

Team CSE4104-7E-T04 | Section 7E | Course: CSE4104

Team Name	CSE4104-7E-T04
Section	7E
Duration	14 weeks
GitHub Repository	https://github.com/SakibHasanAnik/Uniclass-AI

1. Planning Overview

This document describes how our team will build UniClass AI over 14 weeks. The work is divided into four parts: frontend, backend, database and AI. Each member owns one part, and the team leader also handles integration. We will first build the core learning management features (login, courses, materials, assignments, marks and attendance), because the AI model depends on this data. After that we will build and integrate the AI risk advisor, and the last weeks are kept for testing, deployment and the final report.

2. Team Roles and Responsibilities

Member	Role	Responsibilities
Sakib Hasan Anik CSE230321575	Team Leader Database and AI	Database design, Django models, demo data, AI model, explanations, LLM message, Git management, weekly integration, final report coordination
Arnab Basu CSE230321361	Backend Developer	Django REST API, login and role checks, file uploads, dashboard and analytics data, API documentation, backend deployment
Sarder Fariha Alam Trishna CSE230321371	Frontend Developer	UI design, React pages for all roles, dashboards and charts, connecting pages to the API, frontend deployment

3. Work Breakdown

Module	Main tasks	Owner
Authentication	Role-based login, separate admin login, token handling, protected pages	Backend + Frontend
Course management	Courses, sections, class schedule, enrollment, materials	Backend + Frontend
Assignments	Create, submit, grade, late detection	Backend + Frontend

Module	Main tasks	Owner
Marks and attendance	Assessments, marks entry with validation, daily attendance	Backend + Frontend
Dashboards and analytics	Student, teacher, advisor and admin dashboards; charts	Backend + Frontend
Database	ER design, models, migrations, seed data, backups	Leader
AI risk model	Features, rule baseline, logistic regression, evaluation, weekly scoring	Leader
AI explanation and message	Reasons, suggested actions, Groq message with fallback	Leader + Backend
Admin console	Users, courses, enrollment, advisor assignment, run AI	Backend + Frontend
Testing and deployment	Role-access tests, bug fixing, Vercel and Render deployment	All

4. Milestones

Milestone	Week	What will be ready
M0: Project setup	2	Proposal, ER diagram, API list, UI designs, GitHub repository
M1: Core LMS	7	Login for all roles; courses, materials, assignments, marks and attendance working
M2: AI advisor	11	Trained model, weekly risk scores with reasons, advisor portal and AI message
M3: Final system	13	Tested and deployed system, draft report
M4: Final delivery	14	Final report, presentation and live demo

5. Initial Repository Structure

The repository is organised so that each member mainly works in their own folder:

Folder	Owner	Contents
frontend/	Sarder Fariha Alam Trishna	React app: pages, components, API calls
backend/	Arnab Basu	Django project with accounts, courses, grades and advising apps (models by the team leader)
database/	Sakib Hasan Anik	ER diagram, table notes, sample SQL, backups

Folder	Owner	Contents
ai/	Sakib Hasan Anik	Feature extraction, model training, scoring, LLM message
docs/	All	API document, report, screenshots
README.md	All	Project description, team information, features, technology stack, how to run

6. Team Collaboration

- **Version control:** each member works on their own branch (frontend-dev, backend-dev, db-ai-dev) and opens a pull request to main. The team leader reviews and merges.
- **Meetings:** one team meeting every week to review progress, run the whole system together and plan the next week.
- **Communication:** a team group chat for daily updates; tasks tracked with GitHub Issues.
- **Shared rules:** the API document is agreed in week 2 and not changed without telling the team; passwords and API keys are never pushed to GitHub.

7. Risks and Mitigation

Risk	Mitigation
No access to real student data	Use realistic synthetic data and the public UCI dataset, and state this clearly in the report
LLM API limit or downtime	Use the LLM only for writing messages, with a template message as fallback
Delay in one part blocks others	API document agreed early; frontend uses sample data until the backend endpoint is ready
Scope too large for the semester	Core features first (Milestone 1); optional extras only if time allows
Integration problems near the end	Weekly full-system test from week 4 onwards

8. Week-by-Week Schedule

Week	Focus	Leader: Sakib (Database + AI)	Backend: Arnab	Frontend: Trishna	Deliverable
1	Project start	Finalise idea and scope; create GitHub repo, folder structure and README; set up PostgreSQL	Set up Python, Django and DRF; study token authentication	Set up Node, Vite, React and Tailwind; study React Router	Proposal, team sheet, repo with initial structure
2	Requirements and design	Draw ER diagram; list all tables and fields; review UI designs	Write API list (endpoints with sample request and response) in docs/api.md	Design all screens (login, admin login, dashboards, course pages)	ER diagram, API document, UI designs
3	Database and project setup	Write Django models for all apps; run migrations	Create Django project with accounts, courses, grades and advising apps; configure CORS and settings	Build login and admin login pages; build common layout (sidebar, top bar)	Database tables created; static login pages
4	Authentication	Write seed data command (demo users, courses, marks, attendance)	Login with role check, admin login, logout, current user endpoint; role permissions	Connect login to API; save token; redirect by role; protect routes	Working login for all four roles
5	Courses and materials	Review models with team; add any missing fields	Course list, course details, enrollment, materials upload and download	Teacher: My Courses and course page (materials tab); Student: My Courses	Teachers upload, students download materials
6	Assignments	Start AI practice on UCI dataset in Jupyter	Assignments, submissions and grading endpoints	Assignment pages for teacher (create, grade) and student (submit)	Assignment flow working end to end
7	Marks and attendance	Test data quality; fix seed data	Assessments, marks save with validation, attendance save, student results	Marks entry table, attendance page, student results and attendance tabs	Milestone 1: core LMS demo
8	Dashboards and features	Feature extraction code (one row per student); rule-based baseline score	Student and teacher dashboard data; course analytics	Student dashboard, teacher dashboard, analytics charts	Dashboards with real data

Week	Focus	Leader: Sakib (Database + AI)	Backend: Arnab	Frontend: Trishna	Deliverable
9	Model training	Train logistic regression; record recall, precision and F1; compare with baseline	Admin endpoints: users, courses, enrollment, advisor assignment	Admin pages: users, courses, enrollment, advisors	Trained model and metrics; admin console
10	Risk scoring	Scoring command: score, level, reasons and suggested action saved weekly	Advisor endpoints: dashboard, student list, student profile; run-AI endpoint	Advisor dashboard with risk table, filters and badges	Risk scores visible to advisors
11	AI message and interventions	Groq message function with template fallback	Message endpoint; intervention endpoints	Student profile page, AI message box, intervention form and history	Milestone 2: full AI flow working
12	Integration and testing	Run full system weekly; check that roles cannot see others' data	Fix API bugs; complete Postman collection	Loading, empty and error states; UI polish; mobile check for student pages	Tested and stable system
13	Deployment and documentation	Database backup; AI evaluation charts; write AI and database chapters	Deploy backend and database on Render	Deploy frontend on Vercel; take screenshots	Online demo; draft report
14	Final delivery	Final report review; prepare demo script	Help with report and presentation	Prepare presentation slides	Final report, presentation, live demo